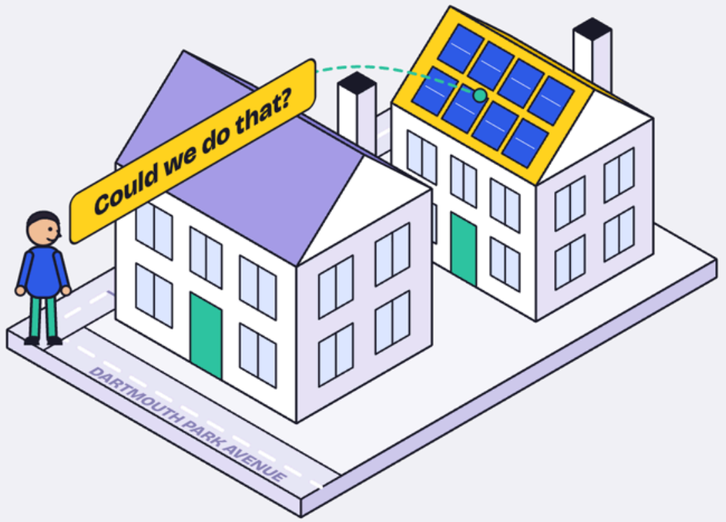
Retrofit Check answers one question for a homeowner in England: can I put solar panels on my house, and where? It also says what they cost and what to do next. Track 1, Solving Fragmented Intelligence, because the answer sits in several public sources that disagree. Built solo.

OPEN AGENT HACKATHON 2026 · TRACK 1

Retrofit Check

Your neighbour gets solar panels. You live on a protected street in London. Are you allowed to install them too, and where on the house?

For homeowners in England on protected streets or in listed buildings. Type an address in, get an answer you can act on.

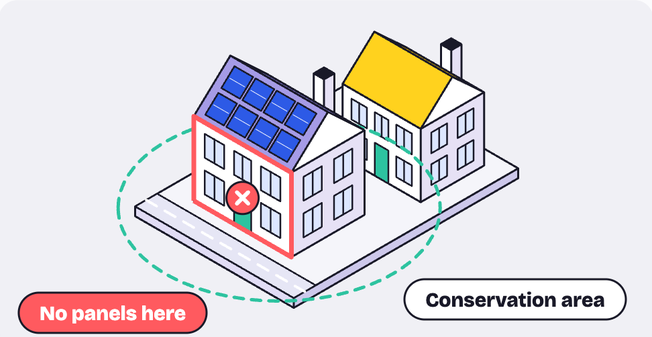


Solving Fragmented Intelligence · Built solo by Aleksandra Vugdragovic

CONTEXT

Usually allowed, with three exceptions.

The usual rule is permitted development: no planning application, as long as the panels sit no higher than the ridge and no more than 20 cm proud of the roof.



Conservation area

A street or district the council protects for its special character.

The roof is fine. No panels on a wall that faces a road (GPDO Part 14 Class A).

9,907 in England (Historic England, 2024). About half of Camden.

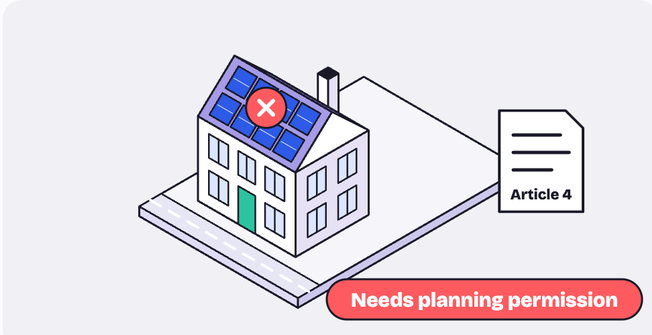


Listed building

A building on the national heritage list: Grade I, II* or II.

Listed building consent first, whatever the usual rule says.

379,443 list entries. One entry can cover a whole terrace.



Article 4 direction

A council order that removes the usual right in an area.

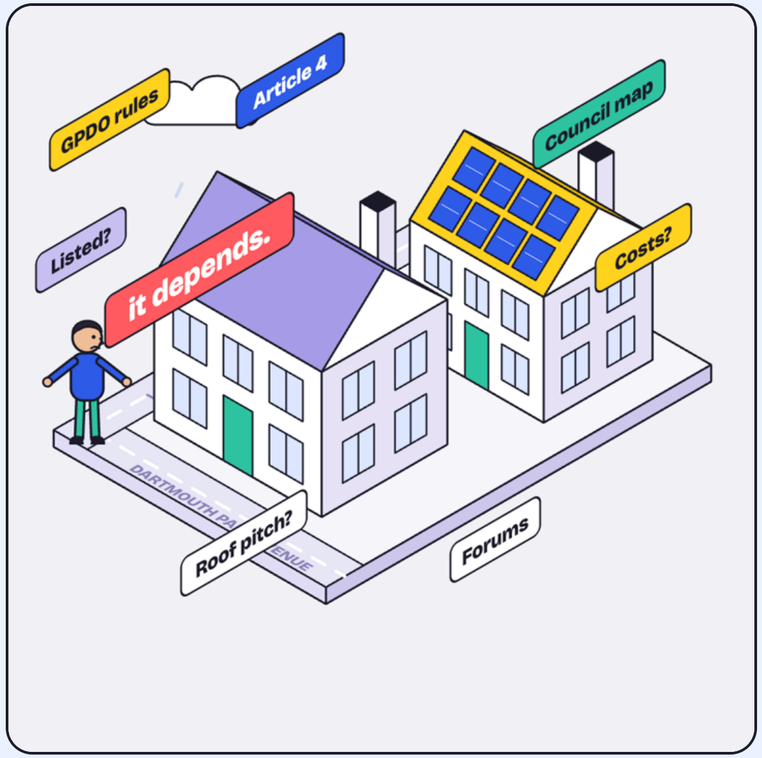
If it covers solar, panels need planning permission.

Camden’s old Hampstead direction covered solar on roofs facing a road, until 9 Jun 2026.

The usual rule is permitted development: solar panels on a house roof need no planning application, as long as they stay below the ridge and no more than 20 cm proud of the roof. Three things change that. In a conservation area, panels may not go on a wall that faces a road. A listed building needs listed building consent first. An Article 4 direction lets a council remove the usual right for an area; Camden’s old Hampstead direction covered solar on roofs facing a road until 9 June 2026. England has 9,907 conservation areas and 379,443 listed building entries (Historic England, 2024).

WHY IT IS HARD

The rules sit in national law, council directions and public records, and they disagree.



A GAP

Haringey has 25 Article 4 areas on planning.data and none for conservation areas. An empty record proves nothing.

A CONFLICT

Camden ended Hampstead’s Article 4 A4Ha1 on 9 June 2026. planning.data still lists it, with no end date.

A GENERAL AI GUESSES

Nemotron on its own got 9 of 15 houses right. **It said listed Keats House could take panels without consent, and never said to ask the council.**

This is the Track 1 problem. The data that answers the question is public, but it is spread over national law, a national database, council pages and other records, and they do not agree. One council publishes none of its conservation area rules to the national database. Another ended a rule the database still lists. A general model asked the same question guesses, sometimes with confidence.

SOLUTION

Address in, answer you can act on.

Retrofit Check Dartmouth Park, NW5 **Find** Rules for England · demo data

RETROFIT CHECK · SOLAR

Can I put solar panels on my house, and where?

Dartmouth Park, NW5

Not listed · Dartmouth Park conservation area, 1992 · south roof 27° · 3,100 kWh a year (you) All 7 facts ›

YOUR HOUSE: 7 FACTS

- Not listed
- Conservation area, 1992
- Article 4: basements
- Road walls: west, south
- South slope 27°
- 2,916 kWh a year
- You use 3,100 kWh

Solar panels

Allowed under permitted development

Not on the west or south wall: they face a road. The roof is fine.

South slope 27° · 2.9 kWp · about 2,916 kWh a year

About 94% of the 3,100 kWh you use a year

£4,600 to £6,100 to install · back in **9 to 22 years**

→ Worth getting quotes **Find MCS certified installers** ↗

Get quotes from two or three MCS certified installers. If you want it in writing, apply to Camden for a lawful development certificate.

Why this answer **More about the money**

This is the answer as the app shows it for house F, copied from the running app on 26 October: the question, the house and its seven facts, and one card. The card gives the status, the condition for this house (no panels on the two walls that face a road), the roof and what it makes, how much of the owner's own use it covers, cost and payback, one recommendation and one action. Other statuses are Needs permission, Needs consent and Check with the council; other recommendations are Ask the council first and Probably not worth it. Why this answer, More about the money, and Sources and limits open on tap, with a source and a date for every fact.

AGENT WORKFLOW

Five agents, two loops.

Planner
Recalls what the owner said, reads planning first, picks what to measure.

Analyst
Reads every house fact through Zetaris.

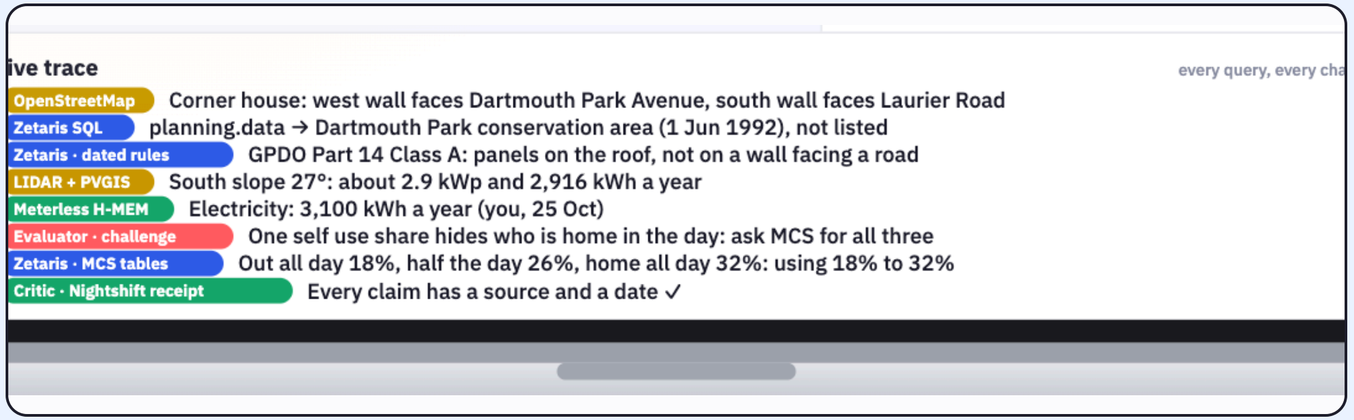
Evaluator
Challenges gaps and conflicts, sends the analyst back.

Reporter
Nemotron writes the answer in plain words.

Critic
Code and Nemotron block any claim without a source.

Loop 1 · Evaluator → Analyst
When sources disagree or a record is missing, the analyst asks Zetaris again.

Loop 2 · Critic → Reporter
A rewrite that adds a number or a place the evidence does not hold is rejected. The checked draft stays.



The evaluator can stop the run and send the analyst back to Zetaris, and the critic can reject the reporter's text. In the trace for house F, the evaluator says one self use share would hide who is home in the day, and the analyst asks the MCS tables for all three cases. Every step is written to logs and shown on the page.

TECHNICAL STACK

Zetaris underneath, agents on top.

Every house fact is a Zetaris query: planning.data and PVGIS as REST tables, the rest in a Postgres datasource.

Every model call runs on NVIDIA Nemotron through Meterless Nightshift. The model is pinned: nvidia/nemotron-3.5-lightning-30b-a3b.

Meterless H-MEM keeps what the owner said, with a date.

Sample mode runs from a fresh clone with no keys.



Zetaris is the only way the agents read house data. The model runs on NVIDIA Nemotron through Meterless Nightshift, and Meterless H-MEM holds what the owner said. Sample mode runs the same agent logic on a cached snapshot, so a judge can run it from a fresh clone without Zetaris, Nightshift or any key. The animation is the stack scene from the project film.

HOW ZETARIS AND METERLESS WERE USED

Every house fact is a Zetaris query. Every model call goes through Nightshift.

Zetaris: the data layer

- planning.data and PVGIS registered as REST tables, queried for each house.
- OpenStreetMap walls, LIDAR roof planes, dated rules, prices and MCS tables in a Postgres datasource.
- The evaluator’s second questions are Zetaris queries too.
- Every query sits in the trace with its purpose, rows and time.

Meterless: model and memory

- Nightshift is the app’s AI provider. Every call logs the model and a receipt.
- NVIDIA Nemotron reads the owner’s note, writes the answer and helps the critic.
- H-MEM keeps the owner’s yearly electricity use with a date. The card cites it as “you, 25 Oct”.
- A correction changes the answer, and a test across two sessions checks it.

Zetaris is load bearing: every house fact the agents use is read through it. OpenStreetMap and LIDAR are fetched once, stored in Postgres and read back through Zetaris. Meterless Nightshift is the only AI provider, and the H-MEM engine powers a feature the owner sees: the card remembers their electricity use and cites it with a date. The H-MEM test runs two sessions and a correction.

WHAT WAS UNIQUE

The agent asks Zetaris a second question before it answers.

Muswell Hill, N10

FIRST QUERY

planning.data, through Zetaris: no Article 4 on this house.

On its own, that reads as Allowed.

SECOND QUERY, FROM THE EVALUATOR

Does Haringey publish conservation area Article 4s at all? 25 Article 4 areas, none for conservation areas.

ANSWER

Check with the council. A missing record proves nothing here.

THE SAME MOVE ON A CONFLICT

Hampstead: planning.data still lists A4Ha1, but Camden's own record ended it on 9 Jun 2026. The agent follows the council.

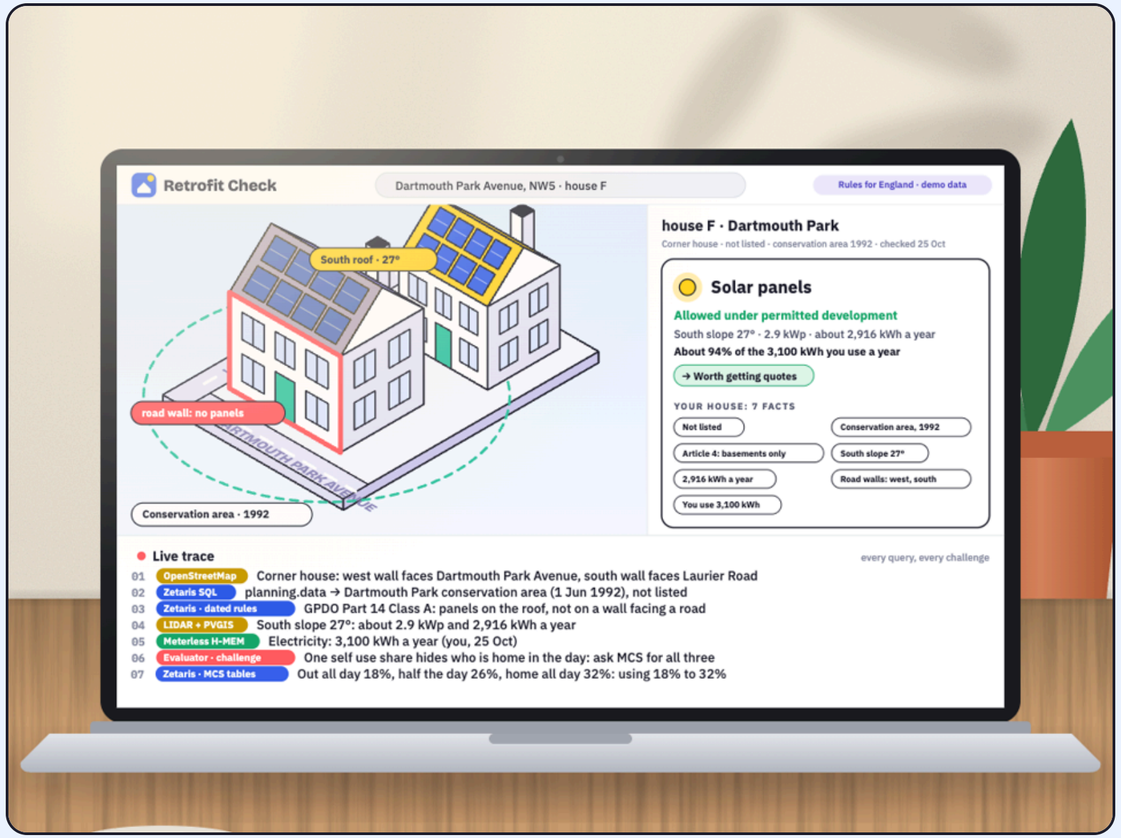
WITHOUT THE EVALUATOR'S CHECKS

The same rules get 13 of 15 houses right. With them, 15 of 15.

Zetaris is not only where the data comes from; the agent also uses it to doubt its own answer. In Muswell Hill the first query finds no Article 4 on the house, which on its own would read as allowed. Before that can stand, the evaluator asks Zetaris a second question: does Haringey publish conservation area Article 4s at all? It has 25 Article 4 areas on planning.data and none for conservation areas, so a missing record proves nothing, and the answer becomes check with the council. In Hampstead the same move settles a conflict: planning.data still lists A4Ha1, Camden ended it on 9 June 2026, and the agent follows the council. Without the evaluator's checks the same rules get 13 of 15 houses right; with them, 15 of 15.

PRODUCT

House F: allowed, but not on the road walls.



South roof, 27°: about 2.9 kWp and 2,916 kWh a year.

£4,600 to £6,100 to install, back in 9 to 22 years.

About 94% of the 3,100 kWh the owner uses.

Not on the west and south walls: they face Dartmouth Park Avenue and Laurier Road.

→ Worth getting quotes

Retrofit Check, house F, as shown in the project film.

House F is a corner house in Dartmouth Park Conservation Area. Two of its walls face roads, so panels may go on the roof but not on those walls. The screen is from the project film; the numbers come from a real run in sample mode, with a yearly electricity use of 3,100 kWh given by the owner.

PROOF

Tested on 15 real London houses against a second source.

System	Right	Confidently wrong	No answer
Retrofit Check	15 of 15	0	0
Same rules, no evaluator	13 of 15	2	0
Nemotron on its own	9 of 15	2	4

Expected answers come from Historic England and were frozen with a hash before any system ran. Fifteen houses is a small test, and it scores the planning status, not the roof.

NOT MEASURED YET

How many owners give up, and how installers check the rules today.

The same rules without the evaluator take planning.data at face value and miss both Hampstead houses. Nemotron on its own, given only the street, postcode district and borough, said listed Keats House needed no consent and never said to ask the council. The test is small and says what it does not cover.

LIMITS AND NEXT

What it does not do yet, and what comes next.

Limits

- Houses only: flats are out of scope.
- Rules for England only. Scotland, Wales and Northern Ireland get a plain “England only” answer.
- What an Article 4 direction removes is on file for Camden and Westminster; anywhere else a house under an Article 4 gets “Check with the council”.
- Houses in a row that OpenStreetMap draws as one building (common for older terraces, for example in Bath) get one outline for several houses: the planning answer still holds, but roof size, yield and money are for the whole row. Houses with their own outline, such as house F, are measured on their own.
- Listing data is incomplete in both sources (planning.data mostly points, Historic England not every outline), so the agent cross checks Historic England and says when it cannot be sure.

Next

- Split a row drawn as one building at the tapped house (for example with HM Land Registry INSPIRE polygons), so the roof numbers are for that house only.
- Heat pump and outside wall insulation answers: rules, grants and prices are already in the database and computed in code, not shown yet.
- Watch this house: tell the owner when a rule, a price or a council decision behind their answer changes.
- Article 4 contents for every council.

Limits first. It answers for houses only; flats are out of scope. The rules are England only, and Scotland, Wales and Northern Ireland get a plain England only answer. What an Article 4 direction removes is on file for Camden and Westminster; elsewhere a house under an Article 4 gets check with the council. A row of houses drawn as one building in OpenStreetMap gets one outline, so the planning answer holds but the roof numbers are for the whole row. Listing data is incomplete in both sources, so the agent cross checks Historic England and says when it cannot be sure. Next: split rows at the tapped house, show the heat pump and outside wall answers that are already computed, watch a house for changes, and add Article 4 contents for every council.